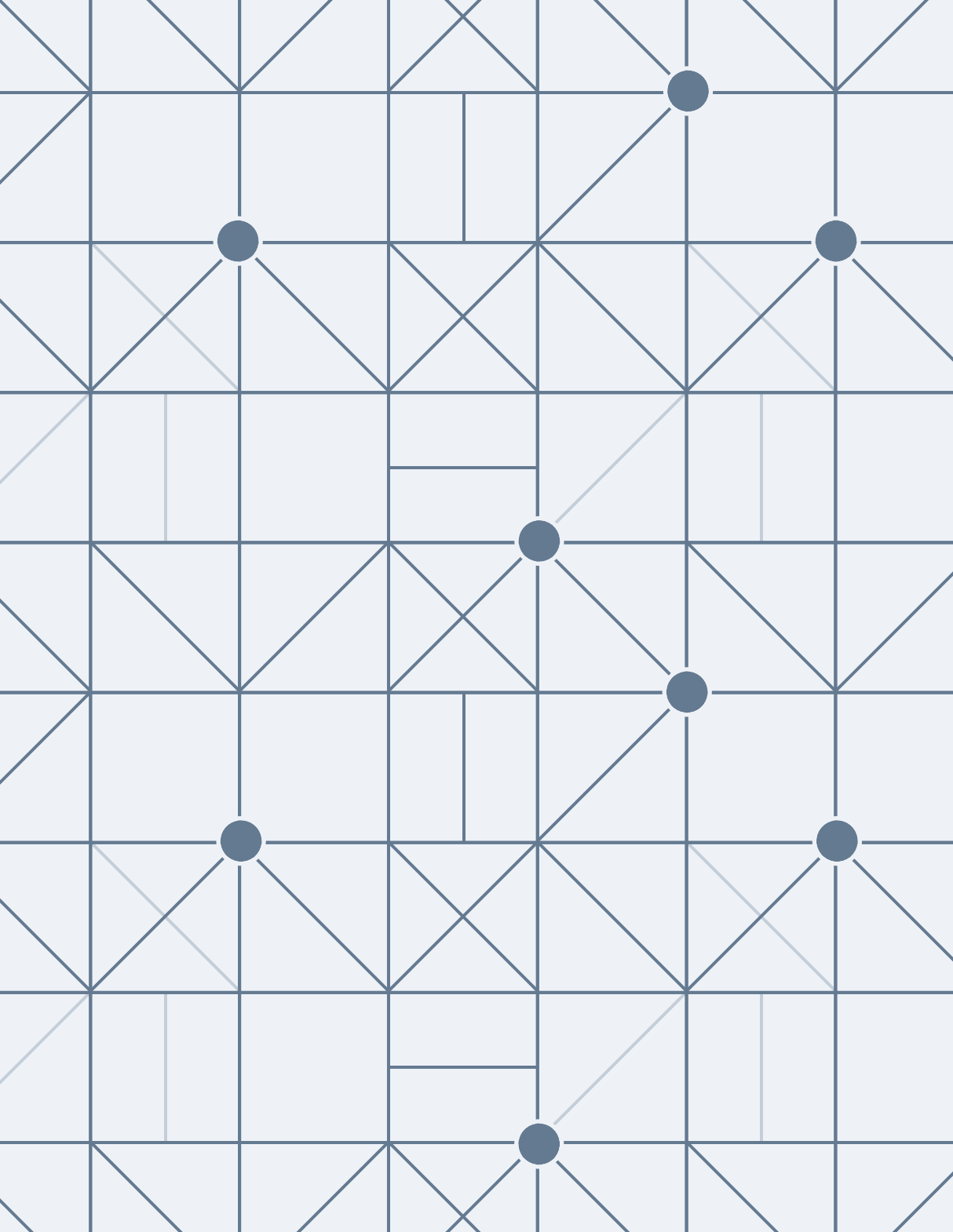
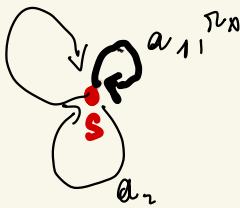




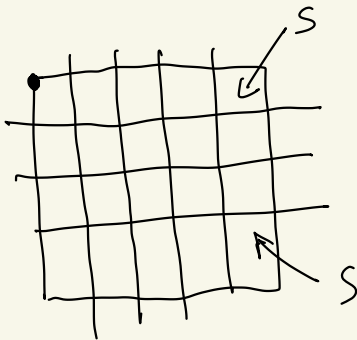


$$Q(a) = \frac{\sum_{i=1}^{t-1} R_i \cdot \mathbb{1}(A_i = a)}{N_{tot}(a)}$$

$$Q_{t+1}(a) = Q_t(a) + \frac{1}{N_{t+1}(a)} \cdot (\underbrace{R_{t+1}}_{\text{reward}} - Q_t(a))$$

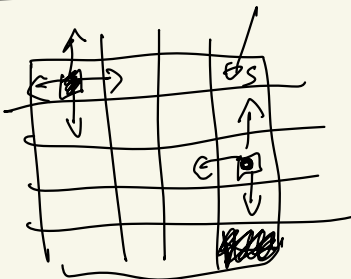
$P(x \in \mathcal{C}) \rightarrow \text{Average A SA}$

$$a = \arg \max_{x \in A} Q_t(x)$$



$$\{ \gamma_1, \dots, \gamma_n \} \rightarrow \gamma_{1,t}$$

$$\{ \gamma_2, \dots, \gamma_n \} \rightarrow$$



$$v_{\pi}(s) = \mathbb{E}(G_T | S_t = s)$$

100	100	100
100	5	100
100	100	100

$$\underline{V_{\pi}(s)} = \mathbb{E} \left[r_{t+1} + \gamma \underline{V_{\pi}(s_{t+1})} \mid s_t = s \right]$$

$$V^*(s) = \max_{a \in A} [r(s, a) + \gamma V^*(s)]$$

✓

- RL INTRODUZIONE

- MULTI-ARMED BANDIT → SOLO AZIONE

- AGGIUNTO LO STATO

- JOB SCHEDULING CON DEADLINE
 - GRIGLIE

→ VALUE FUNCTION



RELAZIONE DI COERENZA



BELLMAN EQUATION



POLICY
ITERATION



VALUE
ITERATION

POLICY → VALUE → POLICY → VALUE

—————→
CONVERGENZA

VALUE ITERATION

$$V^*(s) = \max_{a \in A} R(s, a) + \gamma V(\delta(s, a))$$

PROBLEMA DEVO CONOSCERE LA

DINAMICA DEL SISTEMA, OVVERO
PER OGNI STATO COSA SUCCEDER
SE PRENDENDO UNA AZIONE
SPECIFICA.

SOLUZIONE Q-learning

INTRODOTTO Q PER MULTI-ARMED BANDIT

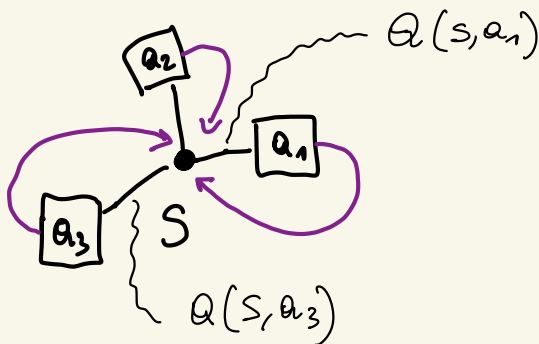
$$Q_t(a) = \frac{\sum_{i=0}^{t-1} R_i \cdot \mathbb{1}(A_i=a)}{N_t(a)}$$

$$Q_{t+1}(a) = Q_t(a) + \frac{1}{N_t(a)} [R_t - Q_t(a)]$$

↑
VECCHIA
STIMA

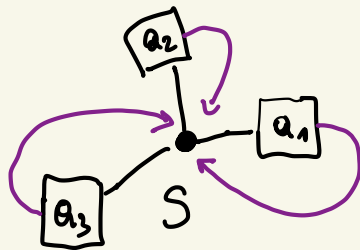
↑
"LEARNING
RATE"

|
ERRORE



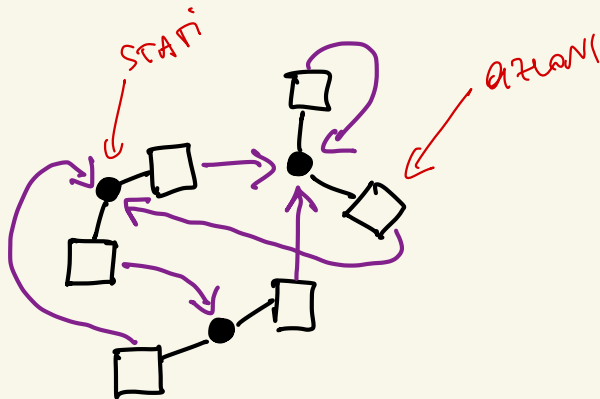
2 CONCETTI → Q VALUES
→ STIMA + LEARNING RATE • ERRORE

SE LO STATO È UNICO
ALLORA È COME SE NON
CI FOSSE!



PROBLEM

E SE L'AZIONE CAMBIASSE
ANCHE LO STATO?



INTRODUZIONE DELLA VALUE-FUNCTION ASSOCIATA A UNA POLICY π

$$V^\pi(s) = \mathbb{E}_\pi (G_t | s_t = s)$$

$$Q^\pi(s, a) = \mathbb{E}_\pi (G_t | s_t = s, a_t = a)$$

+
RELAZIONE DI CONSISTENZA

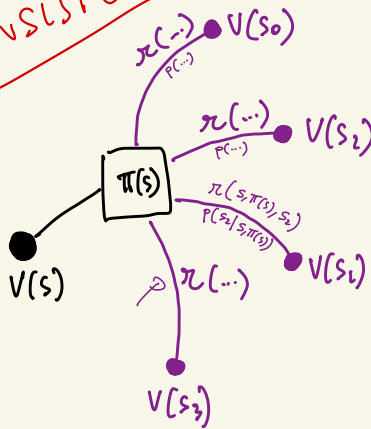
$$\begin{aligned} V^\pi(s) &= \mathbb{E}_\pi (r_{t+1} + \gamma V^\pi(s_{t+1}) | s_t = s) \\ &= \sum_{s'} P(s' | s, \pi(s)) \cdot [r(s, \pi(s), s') + \gamma V^\pi(s')] \end{aligned}$$

+
OTTIMIZZAZIONE

BELLMAN EQUATION

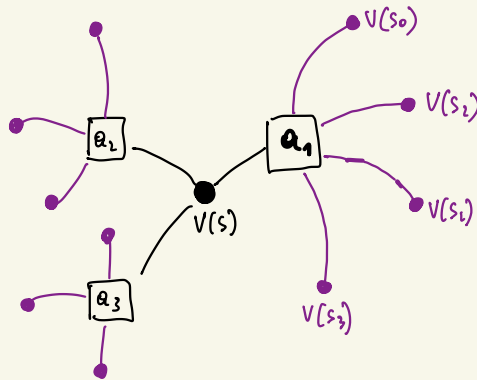
$$V^*(s) = \max_{a \in A} \mathbb{E}_{s' \sim P(\cdot | s, a)} [r(s, a, s') + \gamma V^*(s')]$$

RELAZIONE
DI CONSISTENZA



$$V^\pi(s) = \sum_{s'} P(s' | s, \pi(s)) \cdot [r(s, \pi(s), s') + \gamma V^\pi(s')]$$

BELLMAN EQUATION.



$$V^*(s) = \max_{a \in A} \mathbb{E}_{s' \sim P(\cdot | s, a)} [r(s, a, s') + \gamma V^*(s')]$$

VALUE ITERATION

RISOLVIAMO LA BELLMAN EQUATION

Value Iteration, for estimating $\pi \approx \pi_*$

Algorithm parameter: a small threshold $\theta > 0$ determining accuracy of estimation

Initialize $V(s)$, for all $s \in \mathcal{S}^+$, arbitrarily except that $V(\text{terminal}) = 0$

Loop:

| $\Delta \leftarrow 0$

| Loop for each $s \in \mathcal{S}$:

| $v \leftarrow V(s)$

| $V(s) \leftarrow \max_a \sum_{s',r} p(s',r|s,a) [r + \gamma V(s')]$

| $\Delta \leftarrow \max(\Delta, |v - V(s)|)$

until $\Delta < \theta$

Output a deterministic policy, $\pi \approx \pi_*$, such that

$\pi(s) = \arg \max_a \sum_{s',r} p(s',r|s,a) [r + \gamma V(s')]$

PROBLEMA

$\pi_0 \xrightarrow{s} V \xrightarrow{\pi_0} \pi \xrightarrow{V} \pi^* \xrightarrow{V} V^*$

E SE NON CONOSCO $p(\cdot, \cdot | s, a)$?

OUVERO SE NON CONOSCO IL MODELLO?

DEVO INTERAGIRE CON

L'ENVIRONMENT E QUINDI SIMULARE

ESISTONO 2 APPROCCI

MONTÉ
CARLO

TEMPORAL
DIFFERENCES

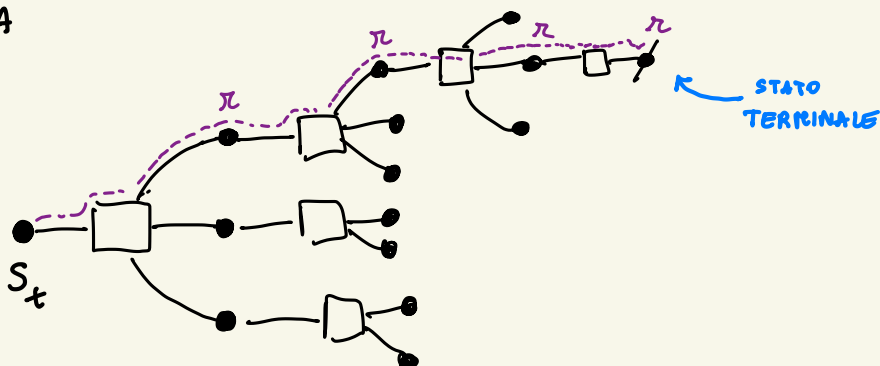
$$V = r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3}$$

$\uparrow$ $\uparrow$ $\uparrow$

$$V = \underline{r_{t+1}} + \boxed{\gamma V}$$

MONTE CARLO APPROACH

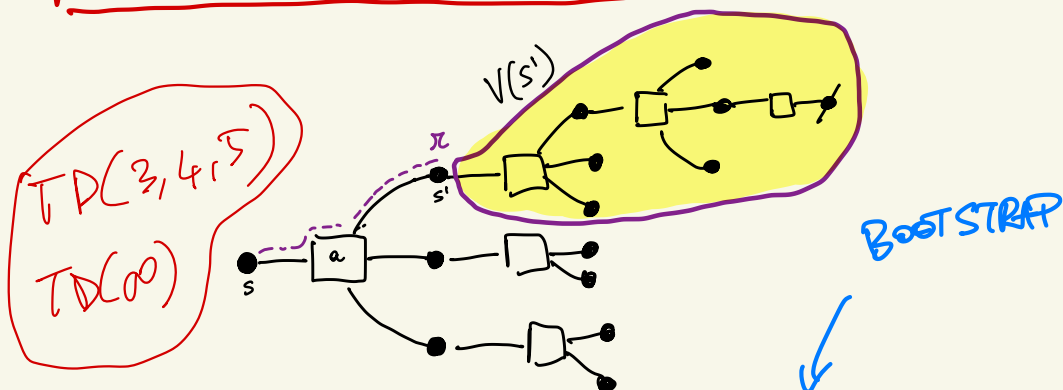
STIMIAMO V^π FACENDO DEI RUN COMPLETI DEL SISTEMA



$$G_t = r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3} + \dots +$$

$$V^\pi(s_t) \leftarrow V^\pi(s_t) + \alpha [G_t - V^\pi(s_t)]$$

TEMPORAL DIFFERENCE APPROACH



$$V^\pi(s_t) \leftarrow V^\pi(s_t) + \alpha [r + \gamma \underbrace{V^\pi(s_{t+1})}_{\text{STIMA DEL RITORNO}} - \underbrace{V^\pi(s_t)}_{\text{VALORE REALE}}]$$

Tabular TD(0) for estimating v_π

Input: the policy π to be evaluated

Algorithm parameter: step size $\alpha \in (0, 1]$

Initialize $V(s)$, for all $s \in \mathcal{S}^+$, arbitrarily except that $V(\text{terminal}) = 0$

Loop for each episode:

Initialize S

Loop for each step of episode:

$A \leftarrow$ action given by π for S

Take action A , observe R, S'

$V(S) \leftarrow V(S) + \alpha[R + \gamma V(S') - V(S)]$

$S \leftarrow S'$

until S is terminal

RIUSCIAMO A STIMARE V^π

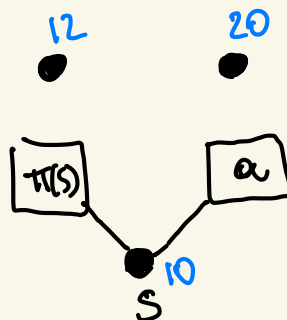
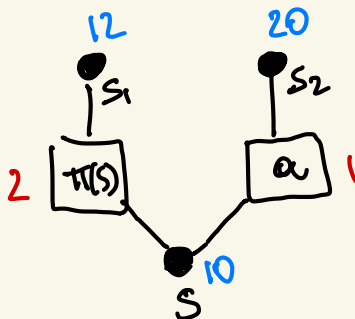
PARTENDO DA UNA POLICY π ANCHE
SE NON CONOSCIAMO IL MODELLO.

RIUSCIAMO PERC' A MIGLIORARE LA POLICY

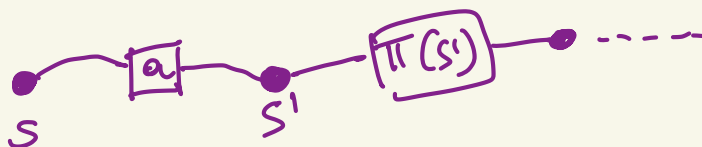
π PARTENDO DA V^π ? E AD IDENTIFI

LA POLICY OTTIMALE π^* ?

↑
POLICY
IMPROVEMENT



MI SERVIREBBE UNA FUNZIONE CHE
 INDIPENDENTEMENTE DAL MODELLO QUANTIFICHI
 IL RITORNO ATTESO A PARTIRE DA s
 PRENDENDO UNA AZIONE a A PIACERE E
 POI SEQUENDO π



$$V^\pi(s) = E_\pi [G_T | S_t = s]$$

$$V^\pi(s) = \sum_{s'} p(s'|s,a) [r(s,a,s') + \gamma V^\pi(s')]$$

$$V^*(s) = \max_{a \in A} \left[\sum_{s'} p(s'|s,a) \cdot [\cdot] \right]$$

$$Q^\pi(s,a) = E_\pi [G_T | S_t = s, A_t = a]$$

$$Q^\pi(s,a) = r(s,a) + \gamma \sum_{s'} p(s'|s,a) \cdot V^\pi(s')$$

$$Q^*(s,a) = r(s,a) + \gamma \sum_{s'} p(s'|s,a) \cdot V^*(s')$$

$$V^*(s) = \max_{a \in A} Q^*(s,a)$$

$$Q^*(s,a) = r(s,a) + \gamma \sum_{s'} p(s'|s,a) \cdot \max_{a' \in A} Q^*(s',a')$$

$$\pi^*(s) = \arg \max_{a \in A} Q^*(s,a)$$

Q-LEARNING

$$Q(s,a) \leftarrow Q(s,a) + \alpha \left[r + \gamma \max_{a'} Q(s',a') - Q(s,a) \right]$$

STIMA DEI
CONTRIBUTI
FUTURI



Q-learning (off-policy TD control) for estimating $\pi \approx \pi_*$

Algorithm parameters: step size $\alpha \in (0, 1]$, small $\varepsilon > 0$

Initialize $Q(s, a)$, for all $s \in \mathcal{S}^+, a \in \mathcal{A}(s)$, arbitrarily except that $Q(\text{terminal}, \cdot) = 0$

Loop for each episode:

Initialize S

Loop for each step of episode:

Choose A from S using policy derived from Q (e.g., ε -greedy)

Take action A , observe R, S'

$Q(S, A) \leftarrow Q(S, A) + \alpha [R + \gamma \max_a Q(S', a) - Q(S, A)]$

$S \leftarrow S'$

until S is terminal

OFF-POLICY

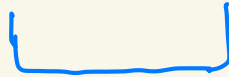
UPDATE DELLA
Q CHE NON
SEGUE LA POLICY
ATTUALE USATA PER
INTERAGIRE CON
L'ENVIRONMENT.

POLICY
PER DECIDERE
NEXT ACTION

CONVERGE A
 π^*

SARSA

$$Q(s,a) \leftarrow Q(s,a) + \alpha \left[r + \gamma Q(s',a') - Q(s,a) \right]$$



STIMA DEI
CONTRIBUTI
FUTURI

Sarsa (on-policy TD control) for estimating $Q \approx q_*$

Algorithm parameters: step size $\alpha \in (0, 1]$, small $\varepsilon > 0$

Initialize $Q(s, a)$, for all $s \in \mathcal{S}^+$, $a \in \mathcal{A}(s)$, arbitrarily except that $Q(\text{terminal}, \cdot) = 0$

Loop for each episode:

Initialize S

Choose A from S using policy derived from Q (e.g., ε -greedy)

Loop for each step of episode:

Take action A , observe R, S'

Choose A' from S' using policy derived from Q (e.g., ε -greedy)

$Q(S, A) \leftarrow Q(S, A) + \alpha [R + \gamma Q(S', A') - Q(S, A)]$

$S \leftarrow S'; A \leftarrow A';$

until S is terminal



ON-POLICY

SI USA UNA POLICY PER PARTE
DUE CASE

- INTERAGIRE CON ENV.
- AGGIORNARE LA Q.

DQN

PROBLEMA: Q-learning lavora con stati o azioni discreti.

e se ho una quantità di stati enorme?
(per esempio immagini)

Q-learning

$$Q(s,a) \leftarrow Q(s,a) + \alpha \left[r + \gamma \max_{a'} Q(s',a') - Q(s,a) \right]$$

scopo è stimare Q^* t.c

$$Q^*(s,a) = r + \gamma \max_{a'} Q(s',a')$$

DQN lo fa usando una rete neurale

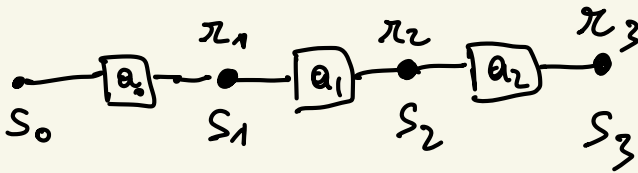
$Q(s,a; \theta)$ che ha come target

$$y(s,a) = r(s,a) + \gamma \max_{a'} Q(s',a'; \theta^-)$$

$$L(\theta) = \mathbb{E} \left[(y - Q(s,a; \theta))^2 \right]$$

APPROCCIO NAIVE (SBAGLIATO)

- INIZIALIZZO LA RETE $Q(\cdot, \cdot; \theta_0)$ CON PARAMETRI θ_0 A CASO.
- ESPLORO BASANDOMI SU QUESTA Q E COLLEZIONO DUE TUPLE.



$$Q(s_0, a_0; \theta_0) \leftarrow \pi_1 + \gamma Q(s_1, a_1; \theta_0)$$

$$\vdots$$

VALORE ATTUALE ———— γ (TARGET)

LOSS

↓
USO IL GRADIENTE PER
OTTIMIZZARE θ_0 DEDUCENDO
COSÌ θ_1

QUESTO APPROCCIO NON CONVERGE
PER 2 RAGIONI

- 1) θ CAMBIA TROPPO VELOCEMENTE
- 2) IL DATO È CORRELATO

APPROCCIO CORRETTO

- INIZIALIZZO LA RETE $Q(\cdot, \cdot; \theta)$
- $\theta^- \leftarrow \theta$
- PER C RASSI ESEGUO QUANTO SEGUE
 - SCELGO AZIONE a con ϵ -greedy su $Q(\cdot, \cdot; \theta)$
 - ESEGUO AZIONE E SALVO LA TUPLA (s, a, z, s') IN UN BUFFER D
 - CAMPIONO UN MINI BATCH CASUALE DA D
 - CALCOLO I TARGET $y = z + \gamma \max_a Q(s, a; \theta^-)$
 $\mathbb{E}((Q(s, a; \theta) - y)^2)$
 - CALCOLO LOSS E FACCO UN PASSO DI GRADIENTE e AGGIORNO I PESI DI θ
- $\theta^- \leftarrow \theta$

APPROCCIO CORRETTO

- INIZIALIZZO LA RETE $Q(\cdot, \cdot; \theta)$
- $\theta^- \leftarrow \theta$
- PER C RASSI ESEGUO QUANTO SEGUE
 - SCELGO AZIONE a con ϵ -greedy su $Q(\cdot, \cdot; \theta)$
 - ESEGUO AZIONE E SALVO LA TUPLA (s, a, r, s') IN UN BUFFER D
 - CAMPIONO UN MINIBATCH CASUALE DA D
 - CALCOLO I TARGET $y = r + \gamma \cancel{\max_a Q(s, a; \theta)}$

$$\mathbb{E}((Q(s, a; \theta) - y)^2)$$
 - CALCOLO LOSS E FACCO UN PASSO DI GRADIENTE E AGGIORNO I PESI DI θ
- $\theta^- \leftarrow \theta$

$\downarrow$
 $Q(s, \max_a Q(s, a; \theta); \theta)$

$\theta^- \leftarrow \theta + \alpha \cdot \nabla_{\theta} Q(s, \max_a Q(s, a; \theta); \theta)$
 $\nabla_a Q(s, a; \theta) \cdot \nabla_{\theta}$

$$Q(s, a)$$

Critic

$$\max_a Q(s, a)$$

$$F(s) \rightarrow a$$

$$F(s, \tilde{\theta}) \quad \underline{\text{ACTOR}}$$

$$F(s, \bar{\theta}) \rightarrow 1$$